

DESHAN FERNANDO

Software Engineer | Backend & Full-Stack

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PROFESSIONAL SUMMARY

Software engineering undergraduate with 10+ months of hands-on internship experience building production backend services at Sri Lanka Telecom. Experienced in designing REST APIs using OOAD principles with a focus on performance (~30% faster endpoints), security (JWT/RBAC), and scalable multi-tenant data isolation. Comfortable working in Agile teams—breaking down user stories, contributing to effort estimates, and communicating progress through demos and written updates. Looking to join a product engineering team as a Software Engineer.

EXPERIENCE

Software Engineer Intern | Sri Lanka Telecom

Jun 2025 - Present | Colombo, Sri Lanka

- Designed and maintained RESTful backend services in Node.js/TypeScript for an enterprise assessment platform serving 5,000+ daily active users.
- Implemented role-based access control (RBAC) and panel-level data isolation, securing 1M+ assessment records across multiple tenant organizations.
- Optimized MySQL queries and added strategic indexes, reducing average API response times by ~30% under production load.
- Developed reusable React components and refactored assessor-facing workflows, improving page load performance and reducing UI bug reports.
- Collaborated in Agile sprints (Jira): clarified user stories and acceptance criteria, contributed to effort estimates, participated in code reviews, and demoed delivered features during sprint reviews.

PROJECTS

LawLinkLK - Client & Lawyer Connecting Platform | lawlinklk.com

- Built and deployed a full-stack web app enabling lawyers and clients to connect via secure real-time messaging, appointment scheduling, and document sharing.
- Engineered the backend with Node.js/Express and MongoDB, implementing JWT authentication, role-based middleware, and Socket.IO for live chat with 50ms latency.
- Integrated an AI-powered legal assistant to help users navigate common legal queries, reducing initial consultation overhead.
- **Tech:** React, Node.js, Express, MongoDB, Socket.IO, Tailwind CSS, JWT

Enterprise Developer API Portal | [GitHub](https://github.com)

- Built a full-stack real-time dashboard aggregating Weather, Stock Market, and Cryptocurrency data from 6+ external APIs under a unified REST interface.
- Implemented enterprise API gateway patterns using WSO2 API Manager (Docker) with JWT validation, OpenAPI 3.0 specs, and versioned endpoints (/api/v1/*).
- Set up CI/CD with GitHub Actions for automated testing and deployment to Render; built multi-API fallback logic to maintain 99%+ demo uptime.
- **Tech:** React, Vite, Node.js, Express, WSO2 API Manager, Docker, GitHub Actions

AI Floorplan-to-3D House Generator

- Designed a multi-stage ML pipeline (object detection → segmentation → geometric post-processing) converting 2D floorplan images into structured 3D-ready geometry.
- Built a Python/FastAPI backend to orchestrate model inference, validate uploads, and cache results; paired with a React (Vite) frontend for job management and visualization.
- **Tech:** Python, PyTorch, Ultralytics YOLO, FastAPI, React, Vite

E-Commerce Microservices Platform

- Architected a scalable backend with Java Spring Boot, decomposing a monolith into RESTful microservices for product catalog, user identity, and order processing.
- Integrated Redis for sub-millisecond product caching and RabbitMQ for async order event processing; containerized with Docker and deployed on AWS EC2.
- **Tech:** Java, Spring Boot, PostgreSQL, Redis, RabbitMQ, Docker, AWS EC2

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Java, Python, SQL

Backend: Node.js, Express.js, Spring Boot, FastAPI, REST APIs, JWT, RBAC, WebSockets

Frontend: React, Vite, HTML5, CSS3, Tailwind CSS, Next.js

Databases: MySQL, MongoDB, PostgreSQL, Redis

DevOps & Tools: Docker, GitHub Actions (CI/CD), Git, Postman, Jira, Linux, AWS EC2

Practices: Agile/Scrum, OOAD, OOP, System Design (Basics), User Stories, Code Reviews, Unit Testing

EDUCATION

BSc (Hons) Computer Science | University of Westminster / IIT Sri Lanka | Jan 2023 - Present

Foundation Certificate in Higher Education - IT (Distinction) | IIT | Jan 2023 - Sep 2023

CERTIFICATIONS

- Postman API Fundamentals - Student Expert
- Introduction to Databases for Back-End Development - Meta (Coursera)
- Microsoft Azure Fundamentals (AZ-900) - LinkedIn Learning

ACHIEVEMENTS

- Team Leader - IEEE Xtreme Encode, a 6-week competitive programming challenge; placed 10th out of 200+ teams in IEEE Region 10 (Asia-Pacific), Jul-Sep 2024.
- Participant - Hult Prize On-Campus Competition 2026.
- Participant — IEEE Xtreme (2025).